

Unit 29: Definite Integral Substitutions and the Area Between Curves

1. The idea, in plain words

You now have every tool needed to tackle one of the most natural geometric questions in all of calculus: what's the area of the region trapped between two curves?

Picture it like this: if one curve sits above another over some stretch of x -values, the region between them is like a “ribbon” of varying width. At each x -value, the ribbon's height is exactly (top curve) minus (bottom curve). Adding up all those tiny heights across the whole stretch — which is exactly what a definite integral does — gives you the ribbon's total area.

The game plan:

1. Find where the two curves intersect. Set the two functions equal to each other and solve — these intersection points become your limits of integration (unless the problem already gives you specific bounds).
2. Figure out which curve is on top in each region between consecutive intersection points. The easiest way: plug in one test value from that region into both functions and see which one comes out bigger.
3. Set up the integral as $\int (\text{top} - \text{bottom}) dx$ over that region.
4. Watch out for curves that cross in the middle of your interval. If the curves swap which one's on top partway through, you can't just integrate straight through — you need to split the integral at each crossing point, using the correct (top – bottom) setup in each separate piece, and then add all the pieces together.
5. Evaluate each piece using antiderivatives — sometimes needing substitution, exactly like Unit 28.

A useful sanity check: area is always positive. If you ever compute a piece and get a negative number, that's a signal you had top and bottom backward for that particular piece — go back and swap them.

2. Toolbox

Area between two curves (where $f(x) \geq g(x)$ on $[a, b]$):

$$A = \int_a^b [f(x) - g(x)] dx$$

Finding the bounds when they're not given: solve $f(x) = g(x)$ for x — the solutions are your intersection points.

When curves cross inside the interval: split into separate pieces at each crossing point, using the correct top-minus-bottom setup in each piece, then add the (positive) areas together:

$$A = \int_a^c [f(x) - g(x)] dx + \int_c^b [g(x) - f(x)] dx \quad (\text{if the curves swap order at } x = c)$$

Substitution still applies exactly as in Unit 28 whenever the (top – bottom) expression you're integrating has a chain-rule pattern hiding inside it.

3. Common mistakes

- Using the wrong bounds. Always double-check whether the problem gives you specific bounds, or whether you're meant to find them yourself from the intersection points — these are not always the same thing.
- Guessing which curve is on top instead of actually testing a point. Always plug a test value from each sub-region into both functions to be sure.
- Forgetting to split the integral when the curves cross in the middle of the region. If you integrate straight through without splitting, positive and negative contributions can partially cancel out, giving you a wrong (usually too-small) answer instead of the true total area.
- Sign errors in the (top – bottom) subtraction, especially when one or both functions already have negative terms.
- Getting a negative “area” and not catching it as a red flag. A negative result always means the top and bottom were swapped somewhere — go back and fix that piece.

4. Problem Set

Warm-up

1. Find the area between $y = x$ and $y = x^2$ on $[0, 1]$.
2. Find the area between $y = 8 - x^2$ and $y = x^2$.
3. Find the area between $y = x^2$ and $y = 2x$.
4. Find the area between $y = x + 6$ and $y = x^2$.
5. Find the area between $y = 4 - x^2$ and the x-axis.
6. Find the area between $y = x$ and $y = x^3$ on $[0, 1]$.

Standard

7. Find the area between $y = x^2 - 4$ and $y = 4 - x^2$.
8. Find the area between $y = x^3$ and $y = 4x$. (Watch for a crossing in the middle of the region!)
9. Find the area between $y = \sqrt{x}$ and $y = \frac{x}{2}$ on $[0, 4]$.
10. Find the area bounded by $y = 6x(x^2 + 1)^2$, the x-axis, $x = 0$, and $x = 1$.
11. Find the area between $y = x^2 + 2$ and $y = -x + 4$.
12. Find the area between $y = x^3 - x$ and the x-axis on $[-1, 1]$. (This region needs to be split.)

Challenge

13. Find the area of the region bounded by $y = -x^2 + 3x$ and $y = 2x^3 - x^2 - 5x$.
14. Find the area between $y = x^4 - 4x^2$ and the x-axis.
15. Find the area between $y = x^3 - 3x$ and $y = x$. (Watch for a crossing in the middle!)
16. Find the area under $y = x\sqrt{9 - x^2}$ from $x = 0$ to $x = 3$ (using the x-axis as the bottom boundary).
17. Find the area under $y = 3x^2(x^3 + 1)$ from $x = -1$ to $x = 1$ (using the x-axis as the bottom boundary).

Applied

18. Company A's revenue rate is $R_A(t) = 100 + 20t$ dollars/week, and Company B's is $R_B(t) = 80 + 15t$ dollars/week, for t in $[0, 10]$ weeks. Find the total extra cumulative revenue Company A earns over Company B during this period.

19. Tank 1 fills at a rate of $r_1(t) = 5 + 2t$ gal/min, and Tank 2 fills at a rate of $r_2(t) = 3 + t$ gal/min, for t in $[0, 6]$ minutes. Find how much more water Tank 1 has collected than Tank 2 after 6 minutes.
20. Runner A's velocity is $v_A(t) = 8 + 0.5t$ mph, and Runner B's velocity is $v_B(t) = 7 + 0.3t$ mph, for t in $[0, 4]$ hours. Find the lead (in miles) that Runner A gains over Runner B during this time.
21. Business Strategy A generates profit at a rate of $p_A(x) = 50 + 3x$ dollars/week, and Strategy B generates profit at a rate of $p_B(x) = 40 + 2x$ dollars/week, for x in $[0, 20]$ weeks. Find the total additional profit Strategy A generates over Strategy B.